

No. of Functional Features Included in the Solution

Date	15 March 2026
Team ID	
Project Name	Credit Card Approval Prediction using Machine Learning
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Applicant Data Input Form	Web form allowing users to enter income, employment, credit history, and other financial/demographic details.	Frontend (Flask templates)	Done	0.75
2	Data Preprocessing & Feature Engineering	Handles missing values and outliers, converts multi-class payment status codes into binary risk labels, encodes categorical variables.	preprocessing.py	Done	0.75
3	Multi-Model Training	Trains four classification algorithms — Logistic Regression, Decision Tree, Random Forest, and XGBoost — on the prepared dataset.	model_training.py	Done	0.75
4	Model Evaluation & Selection	Compares all trained models using accuracy, precision, recall, and F1-score, and selects the best-performing model.	model_evaluation.py	Done	0.5
5	Real-Time Prediction Route	Flask backend route that accepts applicant input, runs it through the saved model, and returns an approval/rejection prediction instantly.	app.py	Done	0.75

6	Prediction Result Display	Displays the predicted approval or rejection outcome to the user on a results page.	result.html	Done	0.5
7	High-Risk Applicant Batch Screening	Allows compliance officers to batch-screen applicants with past-due loan records to flag high-risk, ineligible applicants.	batch_screening.py	In Progress	0.5
8	IBM Watson ML Cloud Deployment	Deploys the selected model to IBM Watson Machine Learning for scalable, cloud-hosted real-time predictions.	watson_deploy.py	Done	0.5

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	7 (1 in progress)
Core / Must-Have Features	6 (Applicant Data Input Form, Preprocessing, Multi-Model Training, Model Evaluation, Prediction Route, Result Display)
Additional / Nice-to-Have Features	2 (High-Risk Batch Screening, IBM Watson Cloud Deployment)
Features Tested & Verified	6

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Applicant Data Input Form; Prediction Result Display
2	Backend / Logic	3	Data Preprocessing & Feature Engineering; Multi-Model Training; Model Evaluation & Selection
3	Database / Storage	1	Serialized model artifact storage (pickle file) for the best-performing model
4	API / Integration	2	Real-Time Prediction Route (Flask); IBM Watson ML Cloud Deployment
5	Security / Authentication	0	Not implemented in the current version; recommended as a future enhancement